



Limits of Local Algorithms over Sparse Random Graphs

[Extended Abstract]^{*}

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ABSTRACT

Local algorithms on graphs are algorithms that run in parallel on the nodes of a graph to compute some global structural feature of the graph. Such algorithms use only local information available at nodes to determine local aspects of the global structure, while also potentially using some randomness. Research over the years has shown that such algorithms can be surprisingly powerful in terms of computing structures like large independent sets in graphs locally. These algorithms have also been implicitly considered in the work on graph limits, where a conjecture due to Hatami, Lovász and Szegedy [17] implied that local algorithms may be able to compute near-maximum independent sets in (sparse) random d -regular graphs. In this paper we refute this conjecture and show that every independent set produced by local algorithms is smaller than the largest one by a multiplicative factor of at least $1/2 + 1/(2\sqrt{2}) \approx .853$, asymptotically as $d \rightarrow \infty$.

Our result is based on an important clustering phenomena predicted first in the literature on spin glasses, and recently proved rigorously for a variety of constraint satisfaction problems on random graphs. Such properties suggest that the geometry of the solution space can be quite intricate. The specific clustering property, that we prove and apply in this paper shows that typically every two large independent sets in a random graph either have a significant intersection, or have a nearly empty intersection. As a result, large independent sets are clustered according to the proximity to each other. While the clustering property was postulated earlier as an obstruction for the success of local algorithms, such as for example, the Belief Propagation algorithm, our result is the first one where the clustering property is used to formally prove limits on local algorithms.

^{*}A full version of this paper is available as *Electronic Colloquium on Computational Complexity, Technical Report TR 13-055, February 20, 2013*.

[†]Research supported by the NSF grants CMMI-1031332.

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ITCS'14, January 12–14, 2014, Princeton, New Jersey, USA.
Copyright 2014 ACM 978-1-4503-2698-8/14/01 ...\$15.00.
<http://dx.doi.org/10.1145/2554797.2554831>.

Categories and Subject Descriptors

F.1.3 [Complexity Measures and Classes]: Lower Bounds

General Terms

Theory

Keywords

Local algorithms, lower bounds, random graphs

1. INTRODUCTION

Local algorithms are decentralized algorithms that run in parallel on nodes in a network using only information available from local neighborhoods to compute some global function of data that is spread over the network. Local algorithms have been studied in the past in various communities. They arise as natural solution concepts in parallel and distributed computing (see, e.g., [21, 19]). They also lead to efficient sub-linear algorithms — algorithms that run in time significantly less than the length of the input — and [26, 25, 16, 27] illustrate some of the progress in this direction. Finally local algorithms have also been proposed as natural heuristics for solving hard optimization problems with the popular Belief Propagation algorithm (see for instance [29, 23]) being one such example. In this work we study the performance of a natural class of local algorithms on *random regular graphs* and show limits on the performance of these algorithms.

1.1 Motivation

The motivation for our work comes from the a notion of local algorithms that has appeared in a completely different mathematical context, namely that of the theory of graph limits, developed in several papers, including [8], [7], [20], [6], [5], [12], [17]. In the realms of this theory it was conjectured that every “reasonable” combinatorial optimization problem on *random graphs* can be solved by means of some local algorithms. To the best of our knowledge this conjecture for the first time was formally stated in Hatami, Lovász and Szegedy in [17, Conjecture 7.13], and thus, from now on, we will refer to it as Hatami-Lovász-Szegedy (or HLS) conjecture, though informally it was posed by Szegedy earlier, and was referenced in several papers, including Lyons and Nazarov [22], and Csoka and Lippner [11]. In the concrete context of the problem of finding largest independent sets in sparse random regular graphs, the conjecture is stated as follows. Let $T_{d,r}$ be a rooted d -regular tree with depth r .

Namely, every node including the root, has degree r , except for the leaves, and the distance from the root to every leaf is r . Consider a function $f_r : [0, 1]^{\mathbb{T}_{d,r}} \rightarrow \{0, 1\}$ which maps every such tree whose nodes are associated with real values from $[0, 1]$ to a “decision” encoded by 0 and 1, where the decision is a function of the values associated with the nodes of the tree. In light of the fact that in a random d -regular graph $\mathbb{G}_d(r)$ the typical node has depth- r neighborhood isomorphic to $\mathbb{T}_{d,r}$, for any constant r , such a function f_r can be used to generate (random) subsets I of $\mathbb{G}_d(r)$ as follows: associate with every node of $\mathbb{G}_d(r)$ a uniform random values from $[0, 1]$ (independently for each node) and apply function f_r to each node. The set of nodes for which f_r produces value 1 defines I , and is called “i.i.d. factor”. It is clear that f_r essentially describes a local algorithm for producing sets I (sweeping issue of computability of f_r under the rug). The HLS conjecture postulates the existence of a sequence of $f_r, r = 1, 2, \dots$, such that the set I thus produced is an independent subset of $\mathbb{G}_d(r)$ and asymptotically achieves the largest possible value as $r \rightarrow \infty$. Namely, largest independent subsets of random regular graphs are i.i.d. factors. The precise connection between this conjecture and the theory of graph limits is beyond the scope of this paper. Instead we refer the reader to the relevant papers [17],[12]¹. The concept of i.i.d. factors appears also in one of the open problem by David Aldous [2] in the context of coding invariant processes on infinite trees.

It turns out that an analogue for the HLS conjecture is indeed valid for another important combinatorial optimization problem - the matching problem. Lyons and Nazarov [22] established it for the case of bi-partite locally $\mathbb{T}_{d,r}$ -tree-like graphs, and Csoka and Lippner established this result for general locally $\mathbb{T}_{d,r}$ -tree-like graphs. Further, one can modify the framework of i.i.d. factors by encapsulating non- $\mathbb{T}_{d,r}$ type neighborhoods, for example by making f_r depend not only on the realization of random uniform in $[0, 1]$ values, but also on the realization of the graph-theoretic neighborhoods around the nodes. Some probabilistic bound on a degree might be needed to make this definition rigorous (though we will not attempt this formalization in this paper). In this case one can consider, for example, i.i.d. factors when neighborhoods are distributed as r generations of a branching process with Poisson distribution, and then ask which combinatorial optimization problems defined now on sparse Erdős-Rényi graphs $\mathbb{G}(n, d/n)$ can be solved as i.i.d. factors. Here $\mathbb{G}(n, d/n)$ is a random graph on n nodes with each of the $\binom{n}{2}$ edges selected with probability d/n , independently for all edges, and $d > 0$ is a fixed constant. In this case it is possible to show that when $c \leq e$, the maximum independent set problem on $\mathbb{G}(n, d/n)$ can be solved nearly optimally by the well known Belief Propagation (BP) algorithm with constantly many rounds. Since the BP is a local algorithm, then the maximum independent set on $\mathbb{G}(n, d/n)$ is an i.i.d. factor, in the extended framework defined above. (We should note that the original proof of Karp and Sipser [18] of the very similar result, relied on a different method.) Thus, the framework of local algorithms viewed as i.i.d. factors is rich enough to solve several interesting combinatorial optimization problems.

¹We also remark that the language used in [17] is quite different from ours. Other works, e.g., [22] however do use language similar to ours in describing this conjecture.

Nevertheless, in this paper we refute the HLS conjecture in the context of maximum independent set problem on random regular graphs $\mathbb{G}_d(n)$. Specifically, we show that for large enough d , with high probability as $n \rightarrow \infty$, every independent set producible as an i.i.d. factor is a multiplicative factor $\gamma < 1$ smaller than the largest independent subset of $\mathbb{G}_d(n)$. We establish that γ is asymptotically at most $\frac{1}{2} + \frac{1}{2\sqrt{2}}$ (though we conjecture that the result holds simply for $\gamma = 1/2$, as we discuss in the body of the paper).

1.2 Techniques

Our result is proved in two steps. First we show that large independent sets in random regular graphs are somewhat clustered. Next we show that solutions of natural algorithms are not clustered. We elaborate on the two parts below.

Clustering is a powerful (though fairly simple to establish in our case) phenomenon associated with the solutions of some combinatorial optimization problems on random graphs. Roughly it states that such solutions are clustered in some natural topology associated with the solution space. (We will describe the precise version we care about shortly.) Such phenomena were first conjectured in the theory of spin glasses and later confirmed by rigorous means. Initially, this clustering property was discussed in terms of so-called overlap structure of the solutions of the Sherrington-Kirkpatrick model [28]. Later works highlighted this phenomenon in the context of random K-SAT problem. In particular, in independent works Achlioptas, Coja-Oghlan and Ricci-Tersenghi [1], and Mezard, Mora and Zecchina [24] proved this phenomenon rigorously. We do not define the random K-SAT problem here and instead refer the reader to the aforementioned papers. What these results state is that in certain regimes, the set of satisfying assignments, with high probability, can be clustered into groups such that two solutions within the same cluster agree on a certain minimum number of variables, while two solutions from different clusters have to disagree on a certain minimum number of variables. In particular, one can identify a certain non-empty interval $[z_1, z_2] \subset [0, 1]$ such that *no* two solutions of the random K-SAT problem agree on precisely z fraction of variables for all $z \in [z_1, z_2]$. One can further show that the onset of clustering property occurs when the density of clauses to variables becomes at least $2^K/K$, while at the same time the formula remains satisfiable with high probability, when the density is below $2^K \log 2$. Interestingly, the known algorithms for finding solutions of random instances of K-SAT problem also stop working around the $2^K/K$ threshold. It was widely conjectured that the onset of the clustering phase is the main obstruction for finding such algorithms. In fact, Coja Oghlan [9] showed that the BP algorithm, which was earlier conjectured to be a good contender for solving the random instances of K-SAT problems, also fails when the density of clauses to variables is at least $2^K \log K/K$, though Coja-Oghlan’s approach does not explicitly rely on the clustering property, and one could argue that the connection between the clustering property and the failure of the BP algorithm is coincidental.

Closer to the topic of this paper, the clustering property was also recently established for independent sets in Erdős-Rényi graphs by Coja-Oghlan and Efthymiou [10]. To describe their result, we first note a well-known fact that the largest independent subset of $\mathbb{G}(n, d/n)$ has size approximately $(2 \log d/d)n$, when d is large (see the next section for

precise details). In [10] it is shown that the set of independent sets of size at least approximately $(\log d/d)n$ (namely those within factor $1/2$ of the optimal), are also clustered. Namely, one can split them into groups such that intersection of two independent sets within a group has a large cardinality, while intersection of two independent sets from different groups has a small cardinality. One should note that algorithms for producing large independent subsets of random graphs also stop short factor $1/2$ of the optimal, both in the case of sparse and in the dense random graph cases, as exhibited by the well-known Karp's open problem regarding independent subsets of $\mathbb{G}(n, 1/2)$ [3].

This is almost the result we need for our analysis with two exceptions. First, we need to establish this clustering property for random regular as opposed Erdős-Rényi graphs. Second, and more significantly, the result in [10] applies to *typical* independent sets and does not rule out the possibility that there exist two independent sets with some "intermediate" intersection cardinality, though the number of such pairs is insignificant compared to the total number of independent sets. For our result we need to show that, without exception, every pair of "large" independent sets has either large or small intersection. We indeed establish this, but at the cost of losing additional factor $1/(2\sqrt{2})$. In particular, we show (see Theorem 2.6) that for large enough d , with high probability as $n \rightarrow \infty$, every two independent subsets of $\mathbb{G}_d(n)$ with cardinality asymptotically $(1 + \beta)(\log d/d)n$, where $1 \geq \beta > \frac{1}{2} + \frac{1}{2\sqrt{2}}$ either have intersection size at least $(1 + z)(\log d/d)n$ or at most $(1 - z)(\log d/d)n$, for some $z < \beta$. The result is established using a straightforward first moment argument: we compute the expected number of pairs of independent sets with intersection lying in the interval $[(1 - z)(\log d/d)n, (1 + z)(\log d/d)n]$, and show that this expectation converges to zero exponentially fast. We remark that even though our conclusion is somewhat stronger than that of previous clustering results, our proof of the clustering phenomenon is extremely simple.

With this result at hand, the refutation of the HLS conjecture is fairly simple to derive. We prove (see Theorem 2.7) that if local algorithms can construct independent sets of size asymptotically $(1 + \beta)(\log d/d)n$, then, by means of a simple coupling construction, we can construct two independent sets with intersection size z for *all* z in the interval $[(1 + \beta)^2(\log d/d)^2n, (1 + \beta)(\log d/d)n]$, clearly violating the clustering property. The additional factor $1/(2\sqrt{2})$ is an artifact of the analysis, and hence we believe that our result holds for all $\beta \in (0, 1]$. Namely, no local algorithm is capable of producing independent sets with size larger than factor $1/2$ of the optimal, asymptotically in d . We note again that this coincides with the barrier for known algorithms. It is noteworthy that our result is the first one where algorithmic hardness derivation relies directly on the geometry of the solution space, viz a vi the clustering phenomena, and thus the connection between algorithmic hardness and clustering property is not coincidental.

The remainder of the paper is structured as follows. We introduce some basic material and the HLS conjecture in the next section. In the same section we state our main theorem — non-validity of the conjecture (Theorem 2.5). We also state two secondary theorems, the first describing the overlap structure of independent sets in random graphs (Theorem 2.6) - the main tool in the proof of our result, and the second describing overlaps that can be found if local al-

gorithms work well (Theorem 2.7). We prove our main theorem easily from the two secondary theorems in Section 3. We prove Theorem 2.7 in Section 4. We omit the proof of Theorem 2.6 from this version.

2. PRELIMINARIES AND MAIN RESULT

For convenience, we repeat here some of the notions and definitions already introduced in the first section.

Basic graph terminology.

All graphs in this paper are understood to be simple undirected graphs. Given a graph $\mathbb{G}$ with node set $V(\mathbb{G})$ and edge set $E(\mathbb{G})$, a subset of nodes $I \subset V(\mathbb{G})$ is an independent set if $(u, v) \notin E(\mathbb{G})$ for all $u, v \in I$. A path between nodes u and v with length r is a sequence of nodes $u_1, \dots, u_{r-1}$ such that $(u, u_1), (u_1, u_2), \dots, (u_{r-1}, v) \in E(\mathbb{G})$. The distance between nodes u and v is the length of the shortest path between them. For every positive integer value r and every node $u \in V(\mathbb{G})$, $B_{\mathbb{G}}(u, r)$ denotes the depth- r neighborhood of u in $\mathbb{G}$. Namely, $B_{\mathbb{G}}(u, r)$ is the subgraph of $\mathbb{G}$ induced by nodes v with distance at most r from u . When $\mathbb{G}$ is clear from context we drop the subscript. The degree of a vertex $u \in V(\mathbb{G})$ is the number of vertices v such that $(u, v) \in E(\mathbb{G})$. The degree of a graph $\mathbb{G}$ is the maximum degree of a vertex of $\mathbb{G}$. A graph $\mathbb{G}$ is d -regular if the degree of every node is d .

Random graph preliminaries.

Given a positive real d , $\mathbb{G}(n, d/n)$ denotes the Erdős-Rényi graph on n nodes $[n] \triangleq \{1, 2, \dots, n\}$, with edge probability d/n . Namely each of the $\binom{n}{2}$ edges of a complete graph on n nodes belongs to $E(\mathbb{G}(n, d/n))$ with probability d/n , independently for all edges. Given a positive integer d , $\mathbb{G}_d(n)$ denotes a graph chosen uniformly at random from the space of all d -regular graphs on n nodes. This definition is meaningful only when nd is an even number, which we assume from now on. Given a positive integer m , let $\mathcal{I}(n, d, m)$ denote the set of all independent sets in $\mathbb{G}(n, d/n)$ with cardinality m . $\mathcal{I}_d(n, m)$ stands for a similar set for the case of random regular graphs. Given integers $0 \leq k \leq m$, let $\mathcal{O}(n, d, m, k)$ denote the set of pairs $I, J \in \mathcal{I}(n, d, m)$ such that $|I \cap J| = k$. The definition of the set $\mathcal{O}_d(n, m, k)$ is similar. The sizes of the sets $\mathcal{O}(n, d, m, k)$ and $\mathcal{O}_d(n, m, k)$, and in particular whether these sets are empty or not, is one of our focuses.

Denote by $\alpha(n, d)$ the size of a largest in cardinality independent subset of $\mathbb{G}(n, d/n)$, normalized by n . Namely,

$$\alpha(n, d) = n^{-1} \max\{m : \mathcal{I}(n, d, m) \neq \emptyset\}.$$

$\alpha_d(n)$ stands for the similar quantity for random regular graphs. It is known that $\alpha(n, d)$ and $\alpha_d(n)$ have deterministic limits as $n \rightarrow \infty$.

THEOREM 2.1. *For every $d \in \mathbb{R}_+$ there exists $\alpha(d)$ such that w.h.p. as $n \rightarrow \infty$,*

$$\alpha(n, d) \rightarrow \alpha(d). \quad (1)$$

Similarly, for every positive integer d there exists α_d such that w.h.p. as $n \rightarrow \infty$

$$\alpha_d(n) \rightarrow \alpha_d. \quad (2)$$

Furthermore

$$\alpha(d) = \frac{2 \log d}{d} (1 - o(1)), \quad (3)$$

$$\alpha_d = \frac{2 \log d}{d} (1 - o(1)), \quad (4)$$

as $d \rightarrow \infty$.

The convergence (1) and (2) were established in Bayati, Gamarnik and Tetali [4]. The limits (3) and (4) follow from much older results by Frieze [13] for the case of Erdős-Rényi graphs and by Frieze and Łuczak [14] for the case of random regular graphs, which established these limits in the $\limsup_n$ and $\liminf_n$ sense. The fallout of these results is that graphs $\mathbb{G}(n, d/n)$ and $\mathbb{G}_d(n)$ have independent sets of size up to approximately $(2 \log d/d)n$, when n and d are large, namely in the doubly asymptotic sense when we first take n to infinity and then d to infinity.

Local graph terminology.

A *decision function* is a measurable function $f = f(u, \mathbb{G}, \mathbf{x})$ where $\mathbb{G}$ is a graph on vertex set $[n]$ for some positive integer n , $u \in [n]$ is a vertex and $\mathbf{x} \in [0, 1]^N$ is a sequence of real numbers for some $N \geq n$ and returns a Boolean value $\{0, 1\}$. A decision function f is said to compute an independent set if for every graph $\mathbb{G}$ and every sequence $\mathbf{x}$ and for every pair $(u, v) \in E(\mathbb{G})$ it is the case that either $f(u, \mathbb{G}, \mathbf{x}) = 0$ or $f(v, \mathbb{G}, \mathbf{x}) = 0$, or both. We refer to such an f as an independence function. For an independence function f , graph $\mathbb{G}$ on vertex set $[n]$ and $\mathbf{x} \in [0, 1]^N$ for $N \geq n$, we let $I_{\mathbb{G}}(f, \mathbf{x})$ denote the independent set of $\mathbb{G}$ returned by f , i.e., $I_{\mathbb{G}}(f, \mathbf{x}) = \{u \in [n] \mid f(u, \mathbb{G}, \mathbf{x}) = 1\}$. We will assume later that X is chosen randomly according to some probability distribution. In this case $I_{\mathbb{G}}(f, \mathbf{x})$ is a randomly chosen independent set in $\mathbb{G}$.

We now define the notion of a “local” decision function, i.e., one whose actions depend only on the local structure of a graph and the local randomness. The definition is a natural one, but we formalize it below for completeness. Let $\mathbb{G}_1$ and $\mathbb{G}_2$ be graphs on vertex sets $[n_1]$ and $[n_2]$ respectively. Let $u_1 \in [n_1]$ and $u_2 \in [n_2]$. We say that $\pi : [n_1] \rightarrow [n_2]$ is an r -local isomorphism mapping u_1 to u_2 if π is a graph isomorphism from $B_{\mathbb{G}_1}(u_1, r)$ to $B_{\mathbb{G}_2}(u_2, r)$ (so in particular it is a bijection from $B_{\mathbb{G}_1}(u_1, r)$ to $B_{\mathbb{G}_2}(u_2, r)$, and further it preserves adjacency within $B_{\mathbb{G}_1}(u_1, r)$ and $B_{\mathbb{G}_2}(u_2, r)$). For $\mathbb{G}_1, \mathbb{G}_2, u_1, u_2$ and an r -local isomorphism π , we say sequences $x^{(1)} \in [0, 1]^{n_1}$ and $x^{(2)} \in [0, 1]^{n_2}$ are r -locally equivalent if for every $v \in B_{\mathbb{G}_1}(u_1, r)$ we have $x_v^{(1)} = x_{\pi(v)}^{(2)}$. Finally we say $f(u, \mathbb{G}, x)$ is an r -local function if for every pair of graphs $\mathbb{G}_1, \mathbb{G}_2$, for every pair of vertices $u_1 \in V(\mathbb{G}_1)$ and $u_2 \in V(\mathbb{G}_2)$, for every r -local isomorphism π mapping u_1 to u_2 and r -locally equivalent sequences $x^{(1)}$ and $x^{(2)}$ we have $f(u_1, \mathbb{G}_1, x^{(1)}) = f(u_2, \mathbb{G}_2, x^{(2)})$. We often use the notation f_r to denote an r -local function.

Let $n_{d,r} \triangleq 1 + d \cdot ((d-1)^r - 1)/(d-2)$ denote the number of vertices in a rooted tree of degree d and depth r . We let $\mathbb{T}_{d,r}$ denote a canonical rooted tree on vertex set $[n_{d,r}]$ with root being 1. For $n \geq n_{d,r}$, $\mathbf{x} \in [0, 1]^n$ and an r -local function f_r , we let $f_r(\mathbf{x})$ denote the quantity $f_r(1, \mathbb{T}_{d,r}, \mathbf{x})$. Let $\mathbf{X}$ be chosen according to a uniform distribution on $[0, 1]^n$. The set subset of nodes $I_{\mathbb{G}_d(n)}(f_r, \mathbf{X})$ is called *i.i.d. factor* produced by the r -local function f_r . As we will see below the $\alpha(f_r) \triangleq \frac{1}{n} \cdot \mathbb{E}_{\mathbf{X}}[|I_{\mathbb{G}_d(n)}(f_r, \mathbf{X})|]$ accurately captures (to within an

additive $o(1)$ factor) the density of an independent returned by an r -local independence function f_r on $\mathbb{G}_d(n)$.

First we recall the following folklore proposition which we will also use often in this paper.

PROPOSITION 2.2. *As $n \rightarrow \infty$, with probability tending to 1 almost all local neighborhoods in $\mathbb{G}_d(n)$ look like a tree. Formally, for every d, r and ϵ , for sufficiently large n ,*

$$\mathbb{P}_{\mathbb{G}_d(n)}(|\{u \in [n] \mid B_{\mathbb{G}_d(n)}(u, r) \not\cong \mathbb{T}_{d,r}\}| \geq \epsilon n) \leq \epsilon.$$

This immediately implies that the expected value of the independent set $I_{\mathbb{G}_d(n)}(f_r, \mathbf{X})$ produced by f_r is $\alpha(f_r)n + o(n)$. In fact the following concentration result holds.

PROPOSITION 2.3. *As $n \rightarrow \infty$, with probability tending to 1 the independent set produced by a r -local function f on $\mathbb{G}_d(n)$ is of size $\alpha(f) \cdot n + o(n)$. Formally, for every d, r, ϵ and every r -local function f , for sufficiently large n ,*

$$\mathbb{P}_{\mathbb{G}_d(n), \mathbf{X} \in [0, 1]^N} (|I_{\mathbb{G}_d(n)}(f_r, \mathbf{X})| - \alpha(f_r)n \geq \epsilon n) \leq \epsilon.$$

PROOF. The proof follows from by the fact that the variance of $|I_{\mathbb{G}_d(n), \mathbf{X}}|$ is $O(n)$ and its expectation is $\alpha(f_r)n + o(n)$, and so the concentration follows by Chebychev’s inequality. The bound on the variance in turn follows from the fact that for every graph $\mathbb{G}$, there are at most $O(n)$ pairs of vertices u and v for which the events $f(u, \mathbb{G}, \mathbf{X})$ and $f(v, \mathbb{G}, \mathbf{X})$ are not independent for random $\mathbf{X}$. Details omitted. $\square$

The Hatami-Lovász-Szegedy Conjecture and our result.

We now turn to describing the Hatami-Lovász-Szegedy (HLS) conjecture and our result. Recall α_d defined by (2). The HLS conjecture can be stated as follows.

CONJECTURE 2.4. *There exists a sequence of r -local independence functions $f_r, r \geq 1$ such that almost surely $I(f_r, n)$ is an independent set in $\mathbb{G}_d(n)$ and $\alpha(f_r) \rightarrow \alpha_d$ as $r \rightarrow \infty$.*

Namely, the conjecture asserts the existence of a local algorithm (r -local independence function f_r) which is capable of producing independent sets in $\mathbb{G}_d(r)$ of cardinality close to the largest that exist. For such an algorithm to be efficient the function $f_r(u, \mathbb{G}, \mathbf{x})$ should also be efficiently computable *uniformly*. Even setting this issue aside, we show that there is a limit on the power of local algorithms to find large independent sets in $\mathbb{G}_d(n)$ and in particular the HLS conjecture does not hold. Let $\hat{\alpha}_d = \sup_r \sup_{f_r} \alpha(f_r)$, where the second supremum is taken over all r -local independence functions f_r .

THEOREM 2.5. *[Main] For every $\epsilon > 0$ and all sufficiently large d ,*

$$\frac{\hat{\alpha}_d}{\alpha_d} \leq \frac{1}{2} + \frac{1}{2\sqrt{2}} + \epsilon.$$

That is, for every $\epsilon > 0$ and for all sufficiently large d , a largest independent set obtainable by r -local functions is at most $\frac{1}{2} + \frac{1}{2\sqrt{2}} + \epsilon$ for all r .

Thus for all large enough d there is a multiplicative gap between $\hat{\alpha}_d$ and the independence ratio α_d . That being said, our result does not rule out that for small d , $\hat{\alpha}_d$ in fact equals α_d , thus leaving the HLS conjecture open in this regime.

The two main ingredients in our proof of Theorem 2.5 both deal with the *overlaps* between independent sets in random regular graphs. Informally, our first result on the size of the overlaps shows that in random graphs the overlaps are not of “intermediate” size — this is formalized in Theorem 2.6. We then show that we can apply any r -local function f_r twice, with coupled randomness, to produce two independent sets of intermediate overlap where the size of the overlap depends on the size of the independent sets found by f_r and the level of coupling. This is formalized in Theorem 2.7. Theorem 2.5 follows immediately by combining the two theorems (and appropriate setting of parameters).

Overlaps in random graphs.

We now state our main theorem about the overlap of large independent sets. We interpret the statement after we make the formal statement.

THEOREM 2.6. *For $\beta \in (1/\sqrt{2}, 1)$ and $0 < z < \sqrt{2\beta^2 - 1} < \beta$ and d , let $s = (1 + \beta)d^{-1} \log d$ and let $K(z)$ denote the set of integers between $\frac{(1-z)n \log d}{d}$ and $\frac{(1+z)n \log d}{d}$. Then, for all large enough d , we have*

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{k \in K(z)} \mathcal{O}(n, d, \lfloor sn \rfloor, k) \neq \emptyset\right) = 0, \quad (5)$$

and

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{k \in K(z)} \mathcal{O}_d(n, \lfloor sn \rfloor, k) \neq \emptyset\right) = 0. \quad (6)$$

In other words, both in the Erdős-Rényi and in the random regular graph models, when $\beta > 1/\sqrt{2}$, and d is large enough, with probability approaching unity as $n \rightarrow \infty$, one cannot find a pair of independent sets I and J with size $\lfloor ns \rfloor$, such that their overlap (intersection) has cardinality at least $\frac{n(1-z) \log d}{d}$ and at most $\frac{n(1+z) \log d}{d}$.

Note that for all $\beta > 1/\sqrt{2}$, there exists z satisfying $0 < z < \sqrt{2\beta^2 - 1}$ and so the theorem is not vacuous in this setting. Furthermore as $\beta \rightarrow 1$, z can be chosen arbitrarily close to 1 making the forbidden overlap region extremely broad. That is, as the size of the independent sets in consideration approaches the maximum possible (namely as $\beta \uparrow 1$), and as $d \rightarrow \infty$, we can take $z \rightarrow 1$. In other words, with probability approaching one, two nearly largest independent sets either overlap almost entirely or almost do not have an intersection. This is the key result for establishing our hardness bounds for existence of local algorithms.

A slightly different version of the first of these results can be found as Lemma 12 in [10]. The latter paper shows that if an independent set I is chosen uniformly at random from the set with size nearly $(1 + \beta)n \log d/d$, then with high probability (with respect to the choice of I), there exists an empty overlap region in the sense described above. In fact, this empty overlap region exists for every $\beta \in (0, 1)$, as opposed to just $1 > \beta > 1/2 + 1/(2\sqrt{2})$ as in our case. Unfortunately, this result cannot be used for our purposes, since this result does not rule out the existence of rare sets I for which no empty overlap exists.

Overlapping from local algorithms.

Next we turn to the formalizing the notion of using a local function f_r twice on coupled randomness to produce overlapping independent sets.

Fix an r -local independence function f_r . Given a vector $\mathbf{X} = (X_u, 1 \leq u \leq n)$ of variables $X_u \in [0, 1]$, recall that

$I_{\mathbb{G}}(f_r, \mathbf{X})$ denotes the independent set of $\mathbb{G}$ given by $u \in I_{\mathbb{G}}(f_r, \mathbf{X})$ if and only if $f_r(u, \mathbb{G}, \mathbf{X}) = 1$.

Recall that $\mathbf{X}$ is chosen according to the uniform distribution on $[0, 1]^n$. Namely, X_u are independent and uniformly distributed over $[0, 1]$. In what follows we consider some joint distributions on pairs of vectors $(\mathbf{X}, \mathbf{Y})$ such that marginal distributions on the vector $\mathbf{X}$ and $\mathbf{Y}$ are uniform on $[0, 1]^n$, though $\mathbf{X}$ and $\mathbf{Y}$ are dependent on each other. The intuition behind the proof of Theorem 2.5 is as follows. Note that if $\mathbf{X} = \mathbf{Y}$ then $I_{\mathbb{G}}(f_r, \mathbf{X}) = I_{\mathbb{G}}(f_r, \mathbf{Y})$. As a result the overlap $I_{\mathbb{G}}(f_r, \mathbf{X}) \cap I_{\mathbb{G}}(f_r, \mathbf{Y})$ between $I_{\mathbb{G}}(f_r, \mathbf{X})$ and $I_{\mathbb{G}}(f_r, \mathbf{Y})$ is $\alpha(f_r)n + o(n)$ in expectation. On the other hand, if $\mathbf{X}$ and $\mathbf{Y}$ are independent, then the overlap between $I_{\mathbb{G}}(f_r, \mathbf{X})$ and $I_{\mathbb{G}}(f_r, \mathbf{Y})$ is $\alpha^2(f_r)n + o(n)$ in expectation, since the decision to pick a vertex u in I is independent for most vertices when $\mathbf{X}$ and $\mathbf{Y}$ are independent. (In particular, note that if the local neighborhood around u is a tree, which according to Proposition 2.2 happens with probability approaching unity, then the two decisions are independent, and $u \in I$ with probability $\alpha(f_r)$.) Our main theorem shows that by coupling the variables, the overlap can be arranged to be of any intermediate size, to within an additive $o(n)$ factor. In particular, if $\alpha(f_r)$ exceeds $\frac{1}{2} + \frac{1}{2\sqrt{2}}$ we will be able to show that the overlap can be arranged to be between the values $\frac{(1-z)n \log d}{d}$ and $\frac{(1+z)n \log d}{d}$, described in Theorem 2.6 which contradicts the statement of this theorem.

THEOREM 2.7. *Fix a positive integer d . For constant r , let $f_r(u, \mathbb{G}, \mathbf{x})$ be an r -local independence function and let $\alpha = \alpha(f_r)$. For every $\gamma \in [\alpha^2, \alpha]$ and $\epsilon > 0$, and for every sufficiently large n , there exists a distribution on variables $(\mathbf{X}, \mathbf{Y}) \in [0, 1]^n \times [0, 1]^n$ such that*

$$\mathbb{P}_{\mathbb{G}_d(n), (\mathbf{X}, \mathbf{Y})}(|I_{\mathbb{G}_d(n)}(f_r, \mathbf{X}) \cap I_{\mathbb{G}_d(n)}(f_r, \mathbf{Y})| \notin (\gamma \pm \epsilon)n) \leq \epsilon.$$

3. PROOF OF THEOREM 2.5

We now show how Theorems 2.6 and 2.7 immediately imply Theorem 2.5.

PROOF OF THEOREM 2.5. Fix an r -local function f_r and let $\alpha = \alpha(f_r)$. Fix $0 < \eta < 1$. We will prove below that for sufficiently large d we have $\alpha/\alpha_d \leq 1/2 + 1/(2\sqrt{2}) + \eta$. The theorem will then follow.

Let $\epsilon = \frac{\eta \log d}{2d}$. By Proposition 2.3 we have that almost surely an independent set returned by f_r on $\mathbb{G}_d(n)$ is of size at least $(\alpha - \epsilon)n$. Furthermore for every $\gamma \in [\alpha^2, \alpha]$ we have, by Theorem 2.7, that $\mathbb{G}_d(n)$ almost surely has two independent sets I and J , with

$$|I|, |J| \geq (\alpha - \epsilon)n \text{ and } |I \cap J| \in [(\gamma - \epsilon)n, (\gamma + \epsilon)n]. \quad (7)$$

Finally, by Theorem 2.1, we have that for sufficiently large d , $|I|, |J| \leq (2d^{-1} \log d)(1 + \eta)n \leq 4d^{-1} \log dn$ and so $\alpha^2 \leq d^{-1} \log d$, allowing us to set $\gamma = d^{-1} / \log d$.

Now we apply Theorem 2.6 with $z = \epsilon d / \log d$ and $\beta > \sqrt{\frac{1+z^2}{2}}$. (Note that for this choice we have $z < 1$ and $z < \sqrt{2\beta^2 - 1} < \beta < 1$. We will also use later the fact that for this choice we have $\beta \leq 1/\sqrt{2} + z = 1/\sqrt{2} + \epsilon d^{-1} \log d$.) Theorem 2.6 asserts that almost surely $\mathbb{G}_d(n)$ has no independent sets of size at least $(1 + \beta)d^{-1} \log dn$ with intersection size in $[(1 - z)d^{-1} \log dn, (1 + z)d^{-1} \log dn]$. Since $|I \cap J| \in [(\gamma - \epsilon)n, (\gamma + \epsilon)n] = [(1 - z)d^{-1} \log dn, (1 + z)d^{-1} \log dn]$, we conclude that $\min\{|I|, |J|\} \leq (1 + \beta)d^{-1} \log dn$. Combining with Equation (7) we get that $(\alpha - \epsilon)n \leq \min\{|I|, |J|\} \leq$

$(1 + \beta)d^{-1} \log dn$ and so $\alpha \leq (1 + \beta)d^{-1} \log d + \epsilon$, which by the given bound on β yields $\alpha \leq (1 + 1/\sqrt{2})d^{-1} \log d + 2\epsilon = (1 + 1/\sqrt{2} + \eta)d^{-1} \log d$. On the other hand we also have $\alpha_d \geq (2 - \eta)d^{-1} \log d$. It follows that $\alpha/\alpha_d \leq 1/2 + 1/2\sqrt{2} + \eta$ as desired.

□

4. PROOF OF THEOREM 2.7

For parameter $p \in [0, 1]$, we define the p -correlated distribution on vectors of random variables $(\mathbf{X}, \mathbf{Y})$ to be the following: Let $\mathbf{X}, \mathbf{Z}$ be independent uniform vectors over $[0, 1]^n$. Now let $Z_u = X_u$ with probability p and Y_u with probability $1 - p$ independently for every $u \in V(G)$.

Let $f(u, \mathbb{G}, \mathbf{x})$ and α be as in the theorem statement. Recall that $f(\mathbf{x}) = f(1, \mathbb{T}_{d,r}, \mathbf{x})$ is the decision of f on the canonical tree of degree d and depth r rooted at the vertex 1. Let $\gamma(p)$ be the probability that $f(\mathbf{X}) = 1$ and $f(\mathbf{Y}) = 1$, for p -correlated variables $(\mathbf{X}, \mathbf{Y})$. As with Proposition 2.3 we have the following.

LEMMA 4.1. *For every $d, r, \epsilon > 0$ and r -local function f , for sufficiently large n we have:*

$$\mathbb{P}(|I_{\mathbb{G}_d(n)}(f, \mathbf{X}) \cap I_{\mathbb{G}_d(n)}(f, \mathbf{Y})| - \gamma(p) \cdot n| \geq \epsilon n) \leq \epsilon,$$

where $(\mathbf{X}, \mathbf{Y})$ are p -correlated distributions on $[0, 1]^n$.

PROOF. By Proposition 2.2 we have that almost surely almost all local neighborhoods are trees and so for most vertices u the probability that u is chosen to be in the independent sets $I(f, \mathbf{X})$ and $I(f, \mathbf{Y})$ is $\gamma(p)$. By linearity of expectations we get that $\mathbb{E}[|I(f, \mathbf{X}) \cap I(f, \mathbf{Y})|] = \gamma(p) \cdot n + o(n)$. Again observing that most local neighborhoods are disjoint we have that the variance of $|I(f, \mathbf{X}) \cap I(f, \mathbf{Y})|$ is $O(n)$. We conclude, by applying the Chebychev bound, that $|I(f, \mathbf{X}) \cap I(f, \mathbf{Y})|$ is concentrated around the expectation and the lemma follows. □

We also note that for $p = 1$ and $p = 0$ the quantity $\gamma(p)$ follow immediately from their definition.

PROPOSITION 4.2. $\gamma(1) = \alpha$ and $\gamma(0) = \alpha^2$.

Now to prove Theorem 2.7 it suffices to prove that for every $\gamma \in [\alpha^2, \alpha]$ there exists a p such that $\gamma(p) = \gamma$. We show this next by showing that $\gamma(p)$ is continuous.

LEMMA 4.3. *For every r , $\gamma(p)$ is a continuous function of p .*

PROOF. Let $(W_u, u \in \mathbb{T}_{d,r})$ be random variables associated with nodes in $\mathbb{T}_{d,r}$, uniformly distributed over $[0, 1]$, which are independent for different u and also independent from X_u and Z_u . We use W_u as generators for the events $Y_u = X_u$ vs $Y_u = Z_u$. In particular, given p , set $Y_u = X_u$ if $W_u \leq p$ and $Y_u = Z_u$ otherwise. This process is exactly the process of setting variables Y_u to X_u and Z_u with probabilities p and $1 - p$ respectively, independently for all nodes u . Now fix any $p_1 < p_2$, and let $\delta < (p_2 - p_1)/d^{r+1}$. We use the notation $f_r(X_u, Z_u, W_u, p)$ to denote the value of f_r when the seed variables realization is $(W_u, u \in \mathbb{T}_{d,r})$, and the threshold value p is used. Namely, $f_r(X_u, Z_u, W_u, p) = f_r(X_u \mathbf{1}\{W_u \leq p\} + Z_u \mathbf{1}\{W_u > p\}, u \in \mathbb{T}_{d,r})$. Here, for ease

of notation, the reference to the tree $\mathbb{T}_{d,r}$ is dropped. Utilizing this notation we have

$$\gamma(p) = \mathbb{P}(f_r(X_u) = f_r(X_u, Z_u, W_u, p) = 1).$$

Manipulating the expressions somewhat we find

$$|\gamma(p_2) - \gamma(p_1)| \leq d^{r+1}(p_2 - p_1).$$

Since r is fixed, the continuity of $\gamma(p)$ is established. □

We are now ready to prove Theorem 2.7.

PROOF OF THEOREM 2.7. Given $\gamma \in [\alpha^2, \alpha]$ by Lemma 4.3 we have that there exists a p such that $\gamma = \gamma(p)$. For this choice of p , let $(\mathbf{X}, \mathbf{Y})$ be a pair of p -correlated distributions. Applying Lemma 4.1 to this choice of p , we get that with probability at least $1 - \epsilon$ we have $|I_{\mathbb{G}_d(n)}(f, \mathbf{X}) \cap I_{\mathbb{G}_d(n)}(f, \mathbf{Y})| \in [(\gamma - \epsilon)n, (\gamma + \epsilon)n]$ as desired. □

5. PROOF OF THEOREM 2.6

We omit the proof of this theorem from this version. Details can be found in the full version [15].

Acknowledgements

The first author wishes to thank Microsoft Research New England for providing exciting and hospitable environment during his visit in 2012 when this work was conducted. The same author also wishes acknowledge the general support of NSF grant CMMI-1031332.

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